

Active Inference ~ Twitter spaces #001 ~ December 9th 2021

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It might be just very lagged, but then how can that happen?

Anyways, doesn't matter.

Okay.

Okay.

Well, it should be like recording, I hope, from this part on at least.

And yeah, welcome.

Thanks for suggesting to do it.

And just feel free to kind of start off however you want.

All right.

Thanks for hosting this and thank you for joining.

So where do I start?

I'm trying to attack a fairly big topic here of connecting essentially the Eastern contemplative traditions with the very Western scientific rational point of view.

So, you know, the good start would be to, I guess, to explain really what Buddhism is and rather than what it appears to be.

Right.

Like ultimately, Buddhism is a technology of mind.

It simply is teaching about the nature of reality.

And the whole structure that grew around it so that it became religion is kind of a vehicle that was necessary to propagate the teaching through the ages.

And it was, you know, it was a structure that was necessary to keep it going, keep it clear, keep it reproduced over, you know, across generations and so on and so forth.

And anybody who understands the teaching gets the key to reality.

And this is the realization of a sort of like a Buddha mind, your realization of your true nature.

And then you sort of get the keys to the kingdom.

You have control over your own reality.

Basically, any reality that you want to experience becomes a choice of the mind.

And it's a very desirable state of mind.

And this is the state of mind that enlightened beings that already are in.

And there are lots of them and, you know, countless, countless, countless Buddhas and Bodhisattvas, but we don't see them because they don't exist in our realm.

And it's just a matter of essentially perception.

And it's a change in perception as a result of experience and as a result of knowledge and experience that's necessary in order to transition into, you know, the full capacity of mind, which is simply that mind generates reality.

Reality, there's no such thing as material reality from the Buddhist perspective.

And I can argue that there is no such thing as material reality from the perspective of, you know, physics.

It's simply, you know, what reality is, it's what we experience as reality, the qualia.

It's simply what the fundamental fabric of the universe, which is the universe itself, the computational kind of fractal that the universe is, you know, appears to itself.

It has to appear as like something.

So it appears as some kind of experience.

And what that appearance is also is kind of arbitrary.

So, you know, realization of the Buddha mind and, you know, entering nirvana is essentially a state of mind where what you experience, at least visually, it's something kind of like looking at an extremely high quality HDR video on an OLED screen at, you know, 120 FPS.

And that's with full control of you creating that reality for yourself in just about as deep capacity as you want, you know, like you have infinite compute capacity to do that.

That's the kind of experience of an enlightened being, you know, from the Buddhist perspective.

And the point of Buddhism is essentially a teaching how to get there.

It's about how to realize that an active inference and the free energy principle is exactly the key to the kingdom.

That's exactly how we get there.

So, you know, the free energy principle, it's there are many, many different ways to summarize, you know, what it states.

But in one useful quote that always comes to my mind is from Frank Herbert's Dune.

It's a proverb of the Ben-Agezerit that be prepared to appreciate what you experience.

And this is essentially, you know, kind of a very contemplative meditative statement of the free energy principle.

You have to be prepared to appreciate.

So you have to have a transcendental knowledge of some sort, a knowledge of something that you might experience.

You may experience something that you have not experienced before.

You know, you feel a new basis for describing your experience.

An accumulation of these bases is simply how we create the model, the generative model of the world.

We essentially, you know, accumulate bases in which we can express our experience, then create a kind of a map of that reality, and then generate all kinds of possibilities out of that map.

And that gives us the new directions of, you know, thinking.

But also it's a new direction of acting in the physical space and so on and so forth.

This principle is just, it just is completely universal.

So that is essentially the essence of the Buddhist teachings, that you have to understand what is the fundamental reality and why it is the way it is.

And the argument there is really that, you know, the Buddhist view of the world is correct simply because the world just cannot be any other way.

It's a consequence of the fact that something exists.

If you assume that something exists, then you get the free energy principle out of it.

Because if something exists, this has to be separate from another thing, right?

Like there has to be a boundary, the Markov blanket.

And the Markov blanket, you know, becomes essentially this fabric that creates categories of experience, the things that we actually experience as separate things.

But ultimately under the hood is just one universe, the mind, the Buddha mind.

And that can be made mathematically precise.

Stephen Wolfram, at the end of 2019, he kicked off the Wolfram Physics Project to find a fundamental theory of physics.

And he essentially put physics on top of the language of computation on hypergraphs, rewrite rules on the hypergraph, which is, you know, a computational model as good as any other.

The combinatory logic, lambda calculus, all these things would be pretty good.

But for Wolfram, this particular four-bialism gave him a really interesting insight into the structure of gravity and how it does relates to quantum theory.

But basically what his model is saying is exactly that reality is entirely virtual.

It's not only that, you know, there is no one physics.

There are infinitely many experiences of infinite possible physics in infinite number of universes that are co-present and interact with each other in some sort of complex ways in such a way that the resulting story as perceived by consciousnesses within the universe, the observing consciousness, is not, you know, causally inconsistent.

And the Markov-blanket in active inference is basically, you know, the description of what happens at the boundary between these computational fractals.

And, you know, as far as I was able to determine, it is, you know, the most useful and most fundamental principle in nature in terms of, like, you know, actively acting in the world, understanding what's going on, acquiring knowledge and all of that.

But, like, meditating on the principle itself is extremely useful as a tool for training the mind to sort of, you know, appreciate what is it that we experience.

And this is where it gets really, really interesting when you start thinking about, like, okay, so what is out there and what is it that I actually experience?

So I can say, for instance, like, okay, I see a nice and stable image of the world where, you know, things have definite shapes and colors and, you know, they appear solid and so on and so forth.

But how is my perception being encoded?

You know, I mean, decoded from the signal that I actually see.

And we actually perceive a very small amount of our vision, right?

It's in color.

Most of it is black and white.

Most of it is blurry.

And yet the image that we see, the image in the mind is a composite, a result of a distributed computation carried out by the brain, right?

And the brain is roughly composed of 150,000 functional units, the cortical columns.

And they vote, so to speak, or produce some sort of average.

And that average is what we experience as a world.

So world, you know, even as experienced by us, is not something that's really definite.

We don't have any way of definitely knowing what's out there on the other side of perception.

We can only say what it appears to be.

And, you know, from the Buddhist perspective, the world that we actually are in is, you know, it is this one thing, the void.

You know, from Hindu perspective, it would be the Brahman, the Supreme Self.

From the Hermetic perspective, that would be the All.

I mean, First Magistris, like, four and a half thousand years ago, he already described the nature of reality.

He said that the universe is mental.

Everything exists in the mind of the All.

And that's, you know, that is the other side.

So it's a kind of like a co-recursive construction.

That the reality appears to be stable to us and appears to be, you know, normal and ordinary.

Just because we are all in the habit of being in that kind of state of mind.

But it only takes a different kind of state of mind that can be actually realized as a result of, you know, training and practice and meditation and visualizations and so on and so forth to actually affect the reality around you.

It's possible.

And, you know, like Alexandra David Nell, for instance, she was a traveler who spent, I think, about 50 years or so in Tibet.

The kind of things she described that people do there, you know, were kind of extraordinary, real magic.

And that's just something quite ordinary over there.

At least, you know, until 1963 or whenever China took over.

They developed that kind of technology, you know, for a long time.

Tibet was a place where in the 13th century, the ruler of Tibet married nine princesses from nine neighboring kingdoms.

And each of them brought with her, you know, Buddhist monks to import the technology into Tibet.

And they essentially, you know, reorganized the entire country to be dedicated to study and understanding all the consequences of,

you know, the lesson of Siddhartha Gautama.

So, you know, active inference, it really is a way to get there.

Like that meditating on this principle and putting it in action and applying it in, you know, in the Buddhist way of life is a sure way to get to enlightenment.

It is not something that is, you know, within the realm of possibility to realize, you know, in soever many lifetimes as it used to be the case.

You know, Buddhism had this rather pessimistic worldview that because of your karmic burden, you may not be able to get to enlightenment in this one lifetime.

But because now we have precise mathematical foundation to understand what is the optimal path, you can

actually do it.

You can just get there within a lifetime.

So, you know, I found it to be a fascinating system of thought.

And it starts to have effects on my life that, you know, things do start to look a little bit science fiction.

Strange things are happening to me.

Like just, you know, luck just somehow finds me more and more.

That's the kind of situation.

So the path kind of gets easier as you keep going because the brain just automatically gears itself.

Once you have the complete understanding of the world that comes with the Buddhist viewpoint, the brain eventually automatically gears itself to analyze all this information.

So that, you know, you start just living your life with extreme level of confidence that you're always doing the right thing.

That you're always going the right way.

That you're always saying the right thing and so on and so forth.

That everything is as it should be.

It just becomes automatic.

It's one of the meanings of nirvana or of liberation.

It doesn't mean that you, you know, become mindless.

It's just like you, you just know as a synthesis of your, of your understanding that what you're doing is simply the right course of action.

And it's a feeling that you, you develop.

Just like, you know, a reinforcement learner would have precise description of its signal of like how well it's doing while learning.

That's the kind of feedback loop.

So build, you know, active inference is all about building all of that.

And essentially, you know, in some sense, mathematically, what, what really means is that you want to become the Markov blanket.

That, that is, that is the name of the game.

You, you just want to be the Markov blanket.

The, the thin veil between, you know, two different virtual realities, essentially.

Your, your own internal one and, you know, the external world.

All right.

All right.

I think that's, that's enough on the, on the connection.

Awesome.

Thanks for sharing it and having it just in one direct place.

So I took a bunch of notes and there's a bunch of places to jump in, but just what would be your goals or what would you like to do?

Yeah.

Um, you know, I just like to talk about these ideas, whatever, if there's anything that you found interesting in this, I can expand on that.

As much as you want.

Basically, you know, there aren't that many people in the field who are interested in looking at the world holistically this way.

That we have to integrate all the thinking in the Western intellectual traditions with all the Eastern contemplative traditions.

So I like to talk about it from, from that perspective.

And, you know, um, all I want to do is to just spread that idea.

That's the, that's why I'm here.

All right, cool.

Well, in the YouTube live chat where anyone can add a comment, I'm also just kind of writing down some key themes and points to return back to.

So we got the full first wash of paint on the mural and then in roughly chronological order, here's a few points we can like return to here a little bit more about your perspective.

And about different world knowledge traditions and then also connected to kind of the active basics.

So very early on, you described Buddhism as a technology and then like mind generating reality.

So the idea of mind generating conscious experience, as well as the more sensory and less consciousness oriented take you provided on visual input, which is like we're experiencing a generative visual model, not a recognition sensory processing model from a retina.

The commonality there is this like generative model.

So I just wanted to hear a little more about the generative model or the generative perspective on consciousness.

How do we know?

What other, what other possibilities are there?

Is Buddhism unique here?

How do different Buddhist groups think about this or what's the history on it?

That's a, that's a, that's a big question, but generally, um, the way to look at analyzing the problem of consciousness is that every culture is correct within, um, their context.

So, um, Buddhism is not, um, it is not a superior point of view.

It's just an excellent one, uh, because it's been designed with a certain, uh, you know, teaching tradition, but there are a lot of metaphoric, more metaphorical points of view, um, that describe the fundamental nature of reality that are also simply correct.

And, and they may appear, uh, contradictory.

Like for instance, you know, um, unification of all religions that essentially all the scriptures, if you, you really listen to the prophets, they say the same thing.

Um, um, um, you know, Islam, Christianity, uh, Judaism, um, any major, any major, uh, monotheistic religion just simply has to have the same kind of structure resulting, um, resulting from it.

And the variations really show in social structures.

Um, so you can think about these things as a, um, what's going on is really with the global consciousness is that the universe is trying to understand itself.

And understand itself in such a way that you not only understand itself and becomes enlightened in a single person, but also transmits that understanding,

that perfect understanding to another being, and so far there hasn't really been a time where the, where there had been two fully enlightened people.

As far as I can tell, um, like this perfect transmission of understanding has never really been happening and different cultures in different religions, you know, they're kind of like different initial conditions in the, in the seller automaton of the, of the, of the, of the God mind, uh, that, that we, we really are.

Um, that's compute, um, uh, uh, what it really means to, to, you know, for the consequences of certain kinds of assertions about the nature of reality and, and the laws that follow.

So what, what, what, what should be the natural law?

There is such a thing as the natural law, but it's, it hasn't been a easy question to figure out what it is and all of kind of, you know, history and everything that happened, um, before the world, before eventually we become all enlightened.

Enlightenment is really just working out what is it for system.

So then we have the set of rules that, yeah, we can, we can definitely agree.

This is, this is the way to go.

So none of these points of view are, um, like every, everybody's point of view is essentially correct.

Their, their experience will lead them, um, if understood how to enlightenment and all of these paths are going to be different because everybody has a path that is suitable for them.

Um, you know, in, in, in, in, in their life experience.

So it's a kind of a system that's designed to be fair.

Um, and, uh, you know, Buddhism is just, um, it's just a particular way of, of looking at it, but, um, a very close way of thinking would be Taoism where they express things more in terms of the energy flow, the energy function.

Right.

Um, um, also quite very reminiscent to like a lot of active inference ideas and the minimization of free energy.

The, the Taoist ideal of a Taoist master is that, you know, the master does nothing and leaves nothing undone.

Um, it's, uh, it's just, uh, it's just a being that's just glides on reality effort, effortless, never really, um, struggling to do anything.

So that is, uh, a beautiful metaphorical statement of exactly the same principles, the fundamental principles of nature.

Um, you know, uh, that, that simply have to be there and everybody understood it in some way.

Um, and until, until we got the internet and we really started exchanging ideas and then really deep into that, it was really hard to see for these, um, metaphorical descriptions because they're very culturally rooted and so on and so forth.

So, you know, it takes a, it takes a, it takes a lot of effort to, to really decipher aloud soon and realize that this

is the same thing as, as, as the free energy principle.

So, uh, all the different ways of knowing are, are, are equally, equally valid.

And that's, uh, I think that's a kind of a cool, that's a, that's a really cool feature of the universe is that, you know, everybody has the, uh, everybody's right.

Uh, once they, once they really, um, just simply start describing what's going on.

Um, and everybody adds, um, a new basis because that's, that's what we really are.

We, we are different, um, axis in, um,

in a basis for understanding, uh, for the universe, understanding itself.

And it's both generative that, you know, adding another consciousness creates this new, um, because it's a new way of seeing, it's a new way of, of doing, it's a new way of, uh, of being and so on and so forth.

So there's a generative access to it, but it's also aspect to it, but it's also perceptual, right?

Like the, I can see things that you may not be able to perceive just simply because they have slightly different random

arrangements of, uh, neurons in my brain.

So I have slightly different generative models that pick up slightly different things from vision.

And, um, there's some wonderful experiments who, which, which show how these generative models kind of work.

Um, there's a paper called, um, diamond diamonds in the rough.

And they, what they've done was kind of wonderful.

They, um, asked participants at some, um, graphics conference to draw a picture of a pair.

I think it was like a still life, just a pair on the table.

And, you know, people drew something that was pretty bad most of the time.

Um, and then they just averaged like 60, 60 or 70 images like that.

And the resulting image was astoundingly good.

Any one image that anybody drew on a little tablet was, was pretty terrible, but the resulting image was extremely good.

So, you know, that's, that's basically, um, how far this goes.

It's, it's, it's, this, this principle is distributed everywhere across the entire nature from a single cell to perspective of a single individual to perspective.

And the individuals in societies to perspective between societies and so on and so forth.

Right.

And active inferences, it provides a way of understanding this at every single level of organization.

Um, you know, how societies, uh, interact with each other, how individuals interact with each other, how, um, you know, tradition, religion, and all these things, how they, how they shape the worldviews that generate the worlds, um, around people.

And, and, and, and, and, you know, you really, um, you really are, you really live in the world that you believe you are in.

Changing your belief literally changes the world around you.

It's, it's, it's very strange, but you know, the, if you can just change your belief, the world around you will

simply adapt.

As a result of the, of the, of the function of the mind.

And that can be known, that can kind of happen automatically once you learn how to do it.

Sorry, I keep raving about this, um, that, that, you know, um, this can become automatic, but this is the exciting part of practicing all of this.

You know, um, as I, as I go on with this, with these kinds of meditations, life gets easier.

Um, day after day, like so much easier, um, that is difficult to, to compare it from like just, just a year ago.

Okay.

Thank you for the awesome answer.

So there's a bunch of points we can always return to, but there's a few great questions from the live chat.

So short question, and then a long question.

Okay.

The short question is from NeuroPixel.

Why is it that only one person at a time can be enlightened?

I don't think that it's, um, one person at a time can be, it's just, it hasn't really happened.

Um, so far that two people, you know, became fully enlightened or maybe, okay.

Perhaps it's the case that there has never really been a fully enlightened person.

Um, Buddha, you know, became awake and he was able to describe the, um, true nature of reality, but he never really passed on the button so far as that there was no next living Buddha.

There was no successor, um, immediately following him.

And the reason why it's so difficult to do that is basically the nature of language and the nature of experience.

And I think Buddha simply did not have the full understanding of what it takes to create the same kind of structure that he arrived at, uh, in someone else's mind.

And the purpose of the, uh, Mahayana Buddhism, the, the great vehicle was to essentially create a structure so that eventually the mind that can receive, um, enlightenment like that is, is, is, is, is going to appear, uh, is going to be ready.

So, you know, fast forward, uh, two and a half thousand years.

Now we have a society where, um, people are being, uh, trained in, uh, all kinds of, um, arts of the mind, um, at the, at the level that is unparalleled.

Uh, I think more people are practicing, um, mindfulness meditation and similar, um, uh, you know, mind practices than ever before.

Um, some amazing, uh, content is available out there and people are slowly turning towards becoming more spiritual, more, uh, contemplative.

And it, it is working that now there are more and more individuals who are becoming awake.

Uh, they, they are becoming aware of the nature of reality.

They're starting to speak out that reality is a little bit stranger than it seems.

And it's a, it's kind of a good thing because we could, it really means that we can make things a lot better very quickly.

None of this has to take a long time, um, for, you know, lots of people to become, um, aware and happy.

Um, and it's, uh, it, it, it is kind of a thermodynamic inevitability that this would, this would have to happen.

Um, and I think, um, you know, it's, it's an interesting question, like why things kind of stagnated after Muhammad, um, Muhammad, right?

Like the, the, he was really the last of the prophets who brought some, some really interesting new, um, way of thinking.

And after that, I think relate it was, it's kind of a issue related to population explosion because there are so many, so many more of us, we kept the construct a little bit too stable.

So it was, you know, harder for some people to, um, look behind the veil of reality that is created by this mind of a society that suddenly became kind of integrated, um, because of the internet, because of television.

Right.

We, we literally started living in a shared image of what the world should look like.

And before that, you know, we had very different ideas of what the world should look like, but the global, uh, communications networks, they, they have sort of taken that, that away.

So I think this is, this is the reason why, you know, these sort of individuals became initially, um, uh, uh, you know, fewer and farther between.

And the, the problem always is to transmit the knowledge to another person.

And it's a, it's an extremely complicated problem.

Um, in, in a human language, it becomes almost impossible unless you come with, um, uh, from the same, uh, you know, intellectual tradition, uh, and, or at least you spend the time to understand the tradition.

So like Buddhism was, um, was a fantastic way of transmitting this information, but it wasn't fully effective.

They were missing, um, they were missing some, you know, key components of, of, of being able to, to sort of explain the nature of reality beyond reasonable doubt.

Right. And, and then language of that is kind of, it had to be, uh, something like computation, right?

It has to be, um, entirely abstract because if you can show that starting from an assumption that something exists and can be in relationship with itself.

And these are the only assumptions, for instance, of the Wolfram model, um, consciousness is an inevitable, uh, construct.

That's the sort of basis of the proof to say that, you know, therefore it cannot really be any, any other way.

Um, than this.

All right.

So some historical factors, a little bit of luck of the draw, a little bit of binomial test.

So just one note and then a second question.

So the note was, um, just maybe in the YouTube live chat or the Twitter thread, you could add the paper link to the pair still life drawing example.

Oh yeah.

And then here's the question from Tucker.

How does the pan enlightenment view that we are all on a path to individual and collective enlightenment think about universal finality or the universe going cold thermodynamically?

That's an excellent question.

Um, uh, I'm going to quote Stephen Wolfram here.

Uh, he said that the thermal death of the universe is a very exciting event and it's not really, um, you know, it's not really, um, you know, it's not really, um, death.

It really is just a face transition.

Um, the, the universe that we exist in, right.

It's, it's, it's undergoing a certain kind of bootstrap process.

And while this process is going on, the energies.

[illegible]